

no_grad#

https://pytorch.org/docs/stable/generated/torch.no_grad.html#torch.no_grad

Description:

Context-manager that disables gradient calculation. Disabling gradient calculation is useful for inference, when you are sure that you will not call `Tensor.backward()`. It will reduce memory consumption for computations that would otherwise have `requires_grad=True`. In this mode, the result of every computation will have `requires_grad=False`, even when the inputs have `requires_grad=True`. There is an exception! All factory functions, or functions that create a new Tensor and take a `requires_grad` kwarg, will NOT be affected by this mode. This context manager is thread local; it will not affect computation in other threads. Also functions as a decorator.

Function Signature

```
class torch.no_grad(orig_func=None)[source]#
```

Example::

Code Examples

```
>>> x = torch.tensor([1.], requires_grad=True)
>>> with torch.no_grad():
...     y = x * 2
>>> y.requires_grad
False
>>> @torch.no_grad()
... def doubler(x):
...     return x * 2
>>> z = doubler(x)
>>> z.requires_grad
False
>>> @torch.no_grad()
... def tripler(x):
...     return x * 3
>>> z = tripler(x)
>>> z.requires_grad
False
>>> # factory function exception
>>> with torch.no_grad():
...     a = torch.nn.Parameter(torch.rand(10))
>>> a.requires_grad
True
```

Notes & Warnings

NOTE

Note No-grad is one of several mechanisms that can enable or disable gradients locally see [Locally disabling gradient computation](#) for more information on how they compare.

NOTE

Note This API does not apply to forward-mode AD. If you want to disable forward AD for a computation, you can unpack your dual tensors.

Detailed Documentation

Documentation

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In this mode, the result of every computation will have `requires_grad=False`, even when the inputs have `requires_grad=True`. There is an exception! All factory functions, or functions that create a new Tensor and take a `requires_grad` kwarg, will NOT be affected by this mode.

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